

Minimal Computing Cosmology 3.0

Finite-Source Integrated Model: From a Generated Address Space to Components, Energy, Gravity and Cosmic Expansion

Authors: Wonsik Choi, Jeongin Choi

Date: 7 October 2026

Status: Six-stage computational integration PASS / integrated scientific model PASS-C

ORCID: Wonsik Choi — 0009-0001-4263-9772

Abstract

Minimal Computing Cosmology (MCC) starts from the hypothesis that the universe need not store an infinitely detailed completed future in advance. Instead, present reality may be executed from finite states, relations, boundaries, residues and resource ledgers. MCC 2.3.2 organized SOURCE, common carrier, dimensional filtering, residue, phenotype and gravity/geometry into a common ownership framework, while the later WRRRA_M implementation specialized that framework to our universe. One major gap remained: the executable arithmetic domain used $N=1,000,000$ as a practical cutoff rather than as an internally generated property.

MCC 3.0 closes that slot with a finite Klein–Zeta SOURCE candidate built from three structures already present in the model. The adopted arithmetic ledger $0.05/0.268/0.682$ gives the exact integer resolution $L=500$; the inherited neutrino harmonic ledger $(0,5,29)$ gives $H=29$; and the number of positive nontrivial Riemann-zeta zeros below the finite boundary $T=2\pi H$ gives $C=70$. Under the declared candidate construction

[$N_U=LHC$,]

the generated finite address window is

[$\boxed{N_U=1,015,000}$.]

This value is not presented as an experimentally measured universal constant. It is a conditional MCC/WRRRA prediction under the declared KZF construction.

Replacing the legacy cutoff with this generated window changes the frozen arithmetic ledger only to $5.000002539\%/26.800001829\%/68.199995632\%$. Reapplying only the same two declared arithmetic calibration freedoms, α and h , restores $5\%/26.8\%/68.2\%$ to numerical precision. The resulting state is propagated through component/phenotype bookkeeping, information load, SI energy, pressure, gravity, reference galaxy rotation, conditional lensing and homogeneous expansion. With the downstream SI coefficients and local-gravity response frozen, the present reference outputs are

[$q_0=-0.5285585894$,]

[$a_T=1.1917875314 \times 10^{-10} \text{ m s}^{-2}$,]

[$v_{\text{ref}}=207.5102419 \text{ km s}^{-1}$,]

[$\theta_{\text{lens}}=0.5355822400''$,]

with a uniform-branch acceleration-transition scale factor

[$a_{\text{trans}}=0.6119598068$.]

The work follows one assessment rule throughout: verification input → WRRRA-specific transformation → output → falsification condition. Fifty-four declared Stage 1–5 checks and eleven Stage-6 cross-stage checks all pass. The integrated computational chain therefore closes, but the scientific model remains PASS-C because the physical necessity of $T=2\pi H$, the independence of L, H, C , the uniqueness of $N_U=LHC$, the physical necessity of the Klein-type closure, the uniqueness of the particle-species decoder, and full spin/flavor/confinement/binding closure remain open.

Preface — From an Exhibition to a Starting Rule

The immediate conceptual origin of MCC 3.0 was not a new astronomical measurement. While viewing KOO JEONG A: OUSSSMOS at Leeum Museum of Art in Seoul, an image emerged of twist distributed throughout an entire form rather than concentrated at one local fold. That intuition was subsequently formalized in the precursor study Finite but Boundaryless: A WRRRA Hybrid Expansion and Accumulated Twist Cosmology with Hidden Gravitational State [1].

That paper did not present a Möbius strip as the literal topology of the universe. A Möbius strip has a boundary and the wrong dimension. Instead, it posed a structural question: if a universe is finite yet does not terminate at an edge, how can its state space close, how can transport return to itself, and what residue can an accumulated twist leave?

The precursor chain was

[$\text{global closure} \rightarrow \text{distributed connection} \rightarrow \text{holonomy} \rightarrow \text{transport residue} \rightarrow \text{twist stress} \rightarrow \text{gravity}$.]

The exhibition is acknowledged as a conceptual trigger, not as scientific evidence. The physical equations, computational rules and falsification conditions remain independent research objects.

MCC 3.0 moves that question one level upstream. A finite and boundaryless universe should not merely have a finite spatial description; its executable SOURCE should also be finite. Earlier WRRRA_M work used a finite domain, but its size was supplied externally. Version 3.0 replaces that remaining external cutoff with an internally generated candidate and follows the resulting state all the way down to energy, gravity and expansion.

1. WRRRA and Minimal Computation

Minimal Computing Cosmology asks whether the universe must contain a completed informational copy of every future event, or whether it can execute the next present from the current state and fixed rules.

MCC adopts the latter as a model hypothesis. WRRRA organizes the execution through

[$\boxed{\text{SOURCE} \rightarrow \text{RELATION/FILTER} \rightarrow \text{RESIDUE} \rightarrow \text{RESOURCE/LEDGER} \rightarrow \text{ACTION/PHENOTYPE}}$].]

SOURCE supplies executable state; RELATION and FILTER determine transport and selection; RESIDUE is history retained in the present; the LEDGER owns conserved or transferred resources; PHENOTYPE is the realized output.

The framework does not require our universe to be uniquely privileged. A fixed finite rule may permit more than one realization, with our universe corresponding to one realized branch. Mathematical infinity remains valid mathematics; MCC 3.0 only refuses to require a physically instantiated infinite SOURCE.

[$\text{mathematical infinity} \neq \text{physically instantiated infinite SOURCE}$].]

2. Why an Infinite SOURCE Is Not Required

WRRRA_M already executed a finite address domain

[$n=2, \dots, 10^6$].]

The problem was that 10^6 was a convenience cutoff. It showed that a finite calculation was possible, but not why the source should have that size.

MCC 3.0 therefore demands a starting rule with six properties: finiteness, address generation, nonterminal closure, compatibility with the frozen filter, survival of existing validation ledgers, and fail-closed ownership of any later failure.

This motivates the finite Klein–Zeta source candidate.

3. Finite Klein–Zeta Generator

The adopted arithmetic ledger

[(0.05,0.268,0.682)

$\left(\frac{25}{500}, \frac{134}{500}, \frac{341}{500} \right)$]

has least common exact denominator

[$\boxed{L=500}$.]

The inherited harmonic ledger

[(0,5,29)]

supplies

[$\boxed{H=29}$.]

With the declared finite phase boundary

[$T_H=2\pi$ $H=182.212373908208$,]

the number of positive nontrivial zeta zeros below the boundary is

[$\boxed{C=70}$.]

The 70th and 71st zero heights are

[$\gamma_{70}=182.20707848436646$,]

[$\gamma_{71}=184.87446784838750$.]

A finite Euler–Maclaurin surrogate with cutoff $M=LH=14,500$ reproduces the standard first-70 zero locations with maximum residual of approximately 5.29×10^{-13} . This is a finite numerical cross-check, not an independent proof that the truncated surrogate itself uniquely generates the full zero set.

A Klein-type quotient is used only as a candidate nonterminal closure, schematically

[$(-T, \theta) \sim (T, -\theta)$.]

It is not a claim that measured spacetime has Klein-bottle topology.

The declared finite-source rule is

[$\boxed{N_U=LHC}$,]

giving

[$\boxed{N_U=1,015,000}$.]

4. Generation of Finite Integer Addresses

Let

$$[0 \leq \ell < L, \quad 0 \leq h < H, \quad 1 \leq z \leq C.]$$

The product state is serialized by

$$[n = 1 + \ell + L[h + H(z-1)].]$$

Thus

$$[1 \leq n \leq 1,015,000.]$$

Label 1 remains an inactive multiplicative/vacuum reference and the executable domain is

$$[\boxed{n=2, \ldots, 1,015,000}.]$$

Integers are therefore used as finite execution addresses rather than as a physically presupposed infinite substrate.

5. Prime/Composite Filter

Address weight is

$$[w_n \propto n^{-\alpha}.]$$

The legacy exponent is

$$[\alpha_0 = 1.8996876950554356.]$$

With the old α_0 held fixed at the new N_U , the ledger becomes

$$[0.05000002538983289, \quad 0.26800001828881304, \quad 0.6819999563213559.]$$

The new finite address counts are:

- primes: 79,608,
- even composites: 507,499,
- odd composites: 427,892.

The same two previously declared calibration freedoms give

$$[\boxed{\alpha_1 = 1.8996877935161325},]$$

$$[\boxed{h_1 = 1.4476744689338703}.]$$

The scalar admission comparison

$$[\boxed{\beta_{\rm eff} = 0.8654567230719301}]$$

is a derived readout, not a third fit.

The common ledger then closes at

[\boxed{0.05/0.268/0.682}.]

6. Residue and Component Structure

The arithmetic filter does not directly name physical particles. It first partitions the address measure into phenotype ϕ , resident nonphenotype D , and return R .

[\text{address} \rightarrow \{\phi, D, R\}.]

Stage 3 uses the actual phase-aware phenotype measure

[$p_n \propto w_n \phi(n)$]

rather than replacing it with a scalar approximation.

The candidate contains 427,892 admitted odd-composite addresses with total phenotype weight 0.05.

Using the inherited 15-prime one-each cyclic assembly and the historical fitted neutron candidate labels {5,47}, the neutron fractions are

- small-first: 0.12947198694,
- large-first: 0.12715941170.

The total-variation distance between the two 15-component number distributions is approximately

[0.20472100116.]

Therefore arithmetic assembly is executable, but order sensitivity prevents a unique particle-species decoder.

7. Particles and Phenotype

The inherited 16-channel charge table remains compatible with the finite-source state. Declared proton-electron, neutron, electron-conjugate and neutral motifs remain exactly charge neutral.

The fully antisymmetric three-quark color state retains unit norm and is annihilated by all eight total color generators within numerical precision.

Using the inherited conditional p/n/e decoder:

- small-first: neutron fraction 0.12947198694, $u/d=1.65641446069$,

- large-first: neutron fraction 0.12715941170, u/d=1.66166981400.

Electric neutrality closes by construction, but baryon/lepton origin is not derived.

The following remain OPEN: full spin-flavor wavefunction, confinement Hamiltonian, binding-energy derivation, unique hadron/species assignment and sea/gluon completion.

8. Information Load and Energy

The cosmic additive ledger contains only

[$\boxed{\phi \oplus D \oplus R}$.]

Address response is

[$g_s(n)$

$1 + \lambda_s \frac{\log n}{\log N_U}$,]

with

[$(\lambda_\phi, \lambda_D, \lambda_R) = (0, 0.25, 0.1)$.]

The finite-source address moments are

[$\boxed{(\mu_\phi, \mu_D, \mu_R)}$

$(0.05000000000000001, 0.278925610976744, 0.687742539854239)$.]

The frozen SI coefficients are

[$\eta_\phi = 7.561582499072265 \times 10^{-10}$,]

[$\eta_D = 7.285761328795971 \times 10^{-10}$,]

[$\eta_R = 7.646102818811323 \times 10^{-10} \setminus \{\rm J, m^{-3}\}$.]

At $V_0 = 1 \text{ m}^3$,

[$E_\phi = 3.78079124953614 \times 10^{-11} \{\rm J\}$,]

[$E_D = 2.03218543006515 \times 10^{-10} \{\rm J\}$,]

[$E_R = 5.25855017259595 \times 10^{-10} \{\rm J\}$,]

and

[$\boxed{E_{\rm total} = 7.668814727614716 \times 10^{-10} \{\rm J\}}$.]

Prime-part and 48-channel descriptions are nested nonadditive views of the phenotype budget. Each view separately sums back to E_0 ; alternative views must never be added as new cosmic energy.

9. Gravity and Twist

The precursor finite-boundaryless work interpreted accumulated spatial twist as a possible owner of hidden gravitational stress [1]. Later WRRRA twist-stress work connected that owner to a dark-mass phenotype and a low-acceleration gravitational response [5].

MCC 3.0 does not add a second fundamental force. The resident D-sector is mapped through the already frozen local response.

With the inherited reference $f_c=0.265$,

$$[a_{T,0} = cH_0 \sqrt{\frac{f_c}{8}},]$$

and the candidate state uses

[a_T

$$a_{T,0} \sqrt{\frac{\rho_D}{u_{\text{crit}} f_c}}.]$$

At $a=1$,

$$[\boxed{a_T = 1.1917875313971119 \times 10^{-10} \text{ m, s}^{-2}}.]$$

The same a_T is used for both rotation and conditional lensing.

10. Galaxy Rotation and Lensing

The frozen test source has mass 6×10^{10} solar masses, Plummer scale 3 kpc, test radius 8.2 kpc, finite lens patch radius 200 kpc and impact parameter 10 kpc.

The direct baryonic reference rotation remains

$$[161.4505105 \text{ km, s}^{-1}].]$$

The finite-source total reference rotation is

$$[\boxed{207.5102419 \text{ km, s}^{-1}}].]$$

The same local gravity response yields conditional finite-patch lensing

$$[\boxed{0.5355822400''}].]$$

Relative shifts from the legacy one-million-address branch are approximately 3.20×10^{-6} for rotation and 7.97×10^{-6} for lensing.

These are model continuity outputs, not new astronomical measurements.

11. Cosmic Expansion

The inherited volume exponents are

[(n_\phi,n_D,n_R)=(0,0,3).]

Thus

[\rho_\phi(a)=E_\phi/a^3, \quad \rho_D(a)=E_D/a^3, \quad \rho_R(a)=E_R.]

Pressure is

[P_\phi=P_D=0, \quad P_R=-\rho_R.]

The homogeneous diagnostics are

[\frac{H(a)}{H_0}

\sqrt{\frac{\rho(a)}{u_{\rm crit}}},]

[q(a)= \frac{1}{2} \frac{\rho(a)+3P(a)}{\rho(a)}.]

	a	density J/m^3	H/H0	q
	0.5	2.4540666613×10^-9	1.7888556123	0.1785814590
	1.0	7.6688147276×10^-10	0.9999913260	-0.5285585894
	2.0	5.5598332420×10^-10	0.8514575347	-0.9187161585

The uniform-branch acceleration transition is

[\boxed{ a_{\rm trans}=0.6119598067968817 }.]

12. Proper Time and Record

The inherited proper-time step is

[\Delta\tau= 1.4813019664158691\times10^{-21}\rm s.]

MCC 3.0 does not identify this with Planck time or an absolute minimum physical time.

For the candidate finite carrier, noncommuting internal loads evolve while total energy remains conserved to a maximum relative drift of approximately

[$\boxed{9.99 \times 10^{-16}}$.]

State trace remains unity within numerical precision.

The framework distinguishes state update and ledger conservation from a full autonomous theory of irreversible physical record formation. Physical apparatus, reset and persistent-record dynamics remain open where not explicitly executed.

13. End-to-End Master Ledger

The integrated chain is

[$\boxed{\text{finite rule} \rightarrow \text{finite addresses} \rightarrow \text{filter} \rightarrow \text{residue} \rightarrow \text{component/phenotype} \rightarrow \text{information load} \rightarrow \text{energy} \rightarrow \text{pressure/gravity} \rightarrow \text{rotation/lensing} \rightarrow \text{expansion}}$.]

Only the ϕ, D, R sector energies are additive. Component and channel descriptions are internal alternative decompositions of E_ϕ , not new resources.

This ownership rule prevents a common failure of unification attempts: explaining the same physical content twice and then counting both descriptions as additional energy.

14. Reproduction of Verified Values

The frozen legacy branch reproduces

[(0.05,0.268,0.682)]

and the reference outputs

[$q_0 = -0.52855$,]

[$v = 207.5109051266 \text{ km s}^{-1}$,]

[$\theta = 0.5355865106''$.]

The finite-source branch gives

[$q_0 = -0.5285585894$,]

[$v = 207.5102418659 \text{ km s}^{-1}$,]

$$[\theta=0.5355822400'']$$

The repository contains 54 declared Stage 1–5 checks and 11 Stage-6 cross-stage checks, for a total of

[\boxed{65}]

passing mathematical, implementation and provenance checks. These are not 65 independent physical experiments.

15. MCC/WRRA Predictions

The finite source size

$$[\boxed{N_U=1,015,000}]$$

is an unmeasured output generated after the declared KZF construction is fixed. Its status is therefore a conditional MCC/WRRA prediction.

The homogeneous acceleration-transition scale

$$[\boxed{a_{\rm trans}=0.6119598068}]$$

is another derived output of the frozen energy/pressure branch.

MCC 2.3.2 also carries the earlier frozen neutrino mass pattern

$[m_1=0,\quad m_2=8.62965\text{ \rm meV},\quad m_3=50.05195\text{ \rm meV},]$

$$[\sum m_\nu = 58.6816 \text{ \rm meV}].$$

MCC 3.0 inherits $H=29$ from that prior ledger but does not claim to have independently rederived the neutrino masses in the present six-stage integration.

16. Open Assumptions and Falsification

The following remain open:

1. necessity of $T=2\pi H$;
2. Cartesian independence of L, H, C ;
3. uniqueness of $N_U=LHC$;
4. physical necessity of the Klein-type quotient;
5. microscopic physical status of the zeta-focus rule;
6. unique Prime-Parts to particle-species mapping;
7. full spin/flux/confinement/binding Hamiltonian;

8. baryon/lepton origin;
9. fully covariant inhomogeneous completion;
10. autonomous physical measurement and persistent-record dynamics.

The integrated model must be revised if the finite source fails its frozen checks, a hidden third arithmetic fit or new SI refit becomes necessary, internal component views are double-counted as energy, rotation and lensing require different tuned gravity responses, proper-time conservation fails, cross-stage handoffs disagree, or an open microscopic rule becomes necessary and cannot be consistently supplied.

Undefined calculations remain OPEN rather than being counted as successes.

17. Conclusion

MCC 3.0 does not attempt to enlarge the model indefinitely. Its purpose is to close one chain that had remained split between an upstream finite-universe intuition and a downstream executable cosmology.

The final structure is

[$\boxed{\text{finite generator} \rightarrow \text{finite universe address space} \rightarrow \text{component/phenotype} \rightarrow \text{energy} \rightarrow \text{gravity} \rightarrow \text{galaxy} \rightarrow \text{cosmic expansion}}$].]

The starting point is no longer “assume an infinite set of integers” and not even “choose a convenient million-address cutoff.” It is a finite candidate generation rule whose output becomes the address domain on which the rest of the model runs.

The six-stage computational integration passes. The integrated scientific model remains PASS-C because the finite-source product law and several microscopic particle claims are still conditional.

The image that began with KOO JEONG A: OUSSSMOS was not evidence. It was a question: if a form can be finite without terminating at an edge, can the universe itself be executed from a finite rule rather than from an infinite prewritten catalog? MCC 3.0 gives a first executable candidate answer.

Reproducibility

From the repository root:

```
python -m pip install -r mcc_3_0/requirements.txt
python mcc_3_0/reproduce_mcc_3_0.py
```

Stage 1 is an immutable provenance freeze. Stages 2–5 are replayed in dependency order and Stage 6 audits cross-stage consistency.

The published Stage-6 repository audit has passed. A fully independent clean-machine replay remains an operational reproduction step to be performed by an external verifier.

References

1. Choi, W. (2026). Finite but Boundaryless: A WRRM Hybrid Expansion and Accumulated Twist Cosmology with Hidden Gravitational State, Version 1.0. GitHub: Wonsik-Choi-janefather/wrra-finite-boundaryless-twist-cosmology. The paper explicitly acknowledges KOO JEONG A: OUSMOS, Leeum Museum of Art, Seoul, as conceptual inspiration rather than scientific evidence.
 2. Choi, W. (2026). Minimal Computing Cosmology 2.3.2: Integrated Model. Zenodo. DOI: 10.5281/zenodo.22733000.
 3. Choi, W.; Choi, J. (2026). WRRM Integrated Upstream–Downstream Model 1.0 r1. Zenodo. DOI: 10.5281/zenodo.23126800.
 4. Choi, W.; Choi, J. (2026). WRRM Finite Klein–Zeta Generation Window v0.1. GitHub: Wonsik-Choi-janefather/wrra-m-0.1/upstream/finite_klein_zeta_window_v0_1.
 5. Choi, W. (2026). Twist Stress Universe and the Mass Phenotype of Dark Matter. Zenodo. DOI: 10.5281/zenodo.22700557.
 6. Choi, W. (2026). WRRM Spatial Twist Selection of Galactic Disk Planes and the Dark Mass Phenotype. Zenodo. DOI: 10.5281/zenodo.23003875.
 7. Choi, W. (2026). WRRM Prime-Parts Rendering: Finite Assembly, Neutrality, Composition and Decay. Zenodo. DOI: 10.5281/zenodo.23128482.
 8. NIST Digital Library of Mathematical Functions. Riemann Zeta Function: zeros and Euler–Maclaurin representations.
 9. mpmath documentation. zetazero.
-

Final assessment

Six-stage execution integration: PASS

MCC 3.0 integrated scientific model: PASS-C

Current finite SOURCE candidate:

[$N_U=1,015,000$]